A Project Report

on

INVENTRON- Inventory Management Web App

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Bachelor of Technology

in

COMPUTER ENGINEERING

by

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CERTIFICATE



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-Nikhil Nandkumar Sonawane,
Rihaab Manzur Wadekar

ABSTRACT

In today's rapidly evolving business environment, efficient inventory management is a critical challenge faced by small and medium enterprises (SMEs). Manual methods are time-consuming, error-prone, and lack real-time visibility. To address these challenges, we have developed Inventron, a web-based inventory management system that aims to simplify and automate inventory operations.

Inventron allows business users to manage products, monitor stock levels, record sales, and maintain supplier and customer data — all through an intuitive and responsive dashboard. Key features include real-time updates, low stock alerts, visual analytics, and secure bcrypt-based user authentication.

Built using a modular architecture and open-source technologies like PHP, MySQL, and Bootstrap, Inventron ensures both scalability and accessibility across devices. This project not only demonstrates the application of theoretical knowledge in software engineering, database management, and web development, but also offers a practical solution to a real-world business problem.

By streamlining inventory workflows, Inventron empowers businesses to make data-driven decisions, reduce operational inefficiencies, and ultimately improve customer satisfaction.

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Chapter 1

Introduction

1.1 Inventron – Inventory Management Web App



Figure 1.1: Inventory management

In today's competitive and fast-paced market environment, businesses are constantly seeking efficient ways to manage their resources, reduce operational costs, and improve customer satisfaction. One of the most critical areas where this efficiency can be realized is inventory management. From small local businesses to large enterprises, the need for a system that can accurately track stock, sales, suppliers, and collaborations is more important than ever before.

1.2 Historical Context

Traditionally, inventory was managed manually using registers or spreadsheets. While feasible for a small number of items, these methods quickly become unreliable as business operations grow. Manual systems are prone to errors such as duplicate entries, misplaced records, and inaccurate stock levels. These problems not only lead to financial losses but can also impact customer trust and business reputation.

With the evolution of digital technology and the increasing penetration of the internet, businesses now have the opportunity to shift from manual to automated inventory management systems. Such systems ensure data accuracy, improve operational transparency, and provide actionable insights in real time.

1.3 Problem Scenario

Distributors Wholesalers: Monitor multi-location inventory.

Pharmacy Shops: Track expiry and batch information.

E-commerce Startups: Control product listings and online sales.

Local Manufacturers: Supervise raw materials and final stock levels

Real-World Example

Suppose a mobile accessories shop wants to keep track of various items like chargers, earphones, and phone cases. Each product has different variants, suppliers, and sales patterns. Instead of relying on handwritten logs, Inventron helps:

- Categorize each item and assign SKUs
- Monitor which items are selling fast
- Set reorder levels for quick replenishment
- Analyze sales by day/week/mo
- Manage collaborations with vendors for consistent supply

All this can be done through an intuitive dashboard, which improves business decision-making and reduces the time spent on inventory handling by more than 50

1.4 Why Inventron?

Unlike many existing solutions that are either too expensive or too complex for small businesses, Inventron provides a lightweight, scalable, and cost-effective solution without compromising on essential features. It's not just a tool—it's a digital assistant for managing your business smartly and efficiently.

Chapter 2

Objective and Problem Statement

2.1 Objective

The primary objective of this project is to design, develop, and deploy a user-friendly, scalable, and efficient web-based Inventory Management System called **Inventron**. The system aims to modernize and simplify how businesses manage their inventory, with a focus on improving operational accuracy, efficiency, and real-time visibility into stock and sales.

The key objectives are as follows:

- To eliminate the challenges associated with manual inventory management through digital automation.
- To create a centralized platform for tracking products, suppliers, customers, and transactions.
- To enable business owners to make informed decisions using real-time inventory data and analytical reports.
- To develop an intuitive user interface that caters to users with minimal technical background.
- To provide essential features such as low-stock alerts, product categorization, sales monitoring, and supplier tracking.
- To support accessibility from any modern web browser, ensuring cross-platform compatibility and eliminating the need for software installations.
- To improve collaboration among business stakeholders through secure access controls.

The system should be cost-effective, modular, and adaptable to various business models, including retail stores, pharmacies, wholesalers, and small manufacturers.

2.2 Problem Statement

Inventory management is a critical part of any business dealing with goods or products. Despite the availability of advanced technologies, many small and medium enterprises (SMEs) continue to rely on outdated methods such as handwritten registers or basic spreadsheet tools. These manual systems are not only time-consuming but also prone to errors, lack integration, and offer no real-time monitoring capabilities.

The major challenges faced in traditional inventory management include:

- **Manual Errors:** Human mistakes during data entry can lead to inaccurate inventory counts, duplication, or loss of records.
- **Stock Imbalances:** Without real-time tracking, businesses often face overstocking (leading to waste) or understocking (leading to missed sales).
- **Lack of Visibility:** Store owners have limited or no access to insights such as fast-moving items, low-performing products, or seasonal trends.
- **Complex Supplier Management:** Managing multiple suppliers, purchase records, and payment histories becomes chaotic without a structured system.
- **Time and Resource Waste:** Manual processes require constant oversight, leading to inefficiency and diverted attention from core business activities.
- **Lack of Collaboration:** Inability to allow multiple users (like staff or co-owners) to safely access and manage inventory data.

As businesses grow, these issues become more pronounced and can have a direct impact on profitability and customer satisfaction. There is a clear gap in the availability of affordable, easy-to-use, and reliable inventory management solutions tailored for SMEs.

Inventron addresses this gap by offering a cloud-based inventory management solution that is accessible, feature-rich, and designed for simplicity. It empowers businesses to:

- Gain real-time visibility into their inventory and sales.
- Avoid losses caused by poor stock control and mismanagement.
- Increase efficiency through automation and streamlined processes.
- Maintain comprehensive records of suppliers, purchases, sales, and stock movements.
- Enable secure multi-user collaboration for better internal coordination.

Through Inventron, we aim to make inventory management less of a burden and more of a strategic asset that enhances business performance and growth.

Chapter 3

Technology Stack

3.1 Introduction

Developing a modern web application like **Inventron** requires the use of various technologies that enable front-end design, back-end logic, data storage, and efficient user interaction. The chosen technology stack ensures that the system is robust, secure, responsive, and scalable for future enhancements.

This chapter provides an overview of the core technologies used during the development of Inventron.

3.2 Front-end Technologies

The front-end refers to the part of the application that users interact with. It plays a crucial role in delivering a smooth and intuitive user experience.

- **HTML5 (HyperText Markup Language):** Used for structuring the content of web pages.
- **CSS3 (Cascading Style Sheets):** Used for styling and visually formatting the application layout, ensuring responsiveness and consistency across devices.
- **JavaScript:** Enables dynamic features such as real-time updates, form validations, and interactive elements.
- **Bootstrap:** A front-end CSS framework used for responsive design and pre-built UI components to speed up development.

3.3 Back-end Technologies

The back-end is responsible for the business logic, data processing, and communication between the client (browser) and server.

- **PHP:** A widely-used server-side scripting language that handles request processing, data operations, session handling, and integration with databases.
- **Apache Server:** A reliable and flexible web server used to deploy the application on local and live environments.

3.4 Database Management

- **MySQL:** An open-source relational database management system (RDBMS) used to store, retrieve, and manage structured data such as product details, stock records, users, sales, and suppliers.
- SQL queries were used for creating tables, inserting and updating data, and generating reports.

3.5 Development Environment

- **XAMPP:** An open-source package that includes Apache, MySQL, PHP, and phpMyAdmin. It was used as the local development environment.
- **Visual Studio Code:** The primary code editor used for writing and managing HTML, CSS, PHP, and JavaScript files.
- **phpMyAdmin:** A web-based interface for managing the MySQL database and executing SQL commands.

3.6 Version Control (Optional)

- **Git:** Used to track code changes during the development cycle.
- **GitHub:** A cloud-based repository where the codebase was stored for backup and collaboration.

3.7 Additional Tools

- **Figma:** Used for UI/UX planning and mockup design before the actual development.
- **Browser Developer Tools:** Used for debugging and inspecting HTML, CSS, and JavaScript in real time.

3.8 Rationale Behind Choosing These Technologies

- All technologies used are open-source and widely supported.
- The stack is lightweight, making the system faster and more efficient for SMEs.
- Easy to maintain, extend, and debug.
- Well-documented, allowing for easy onboarding of future developers or contributors.

Chapter 4

Software and Hardware Requirements

4.1 Introduction

To ensure smooth development, deployment, and operation of the Inventron web application, a specific set of software and hardware resources were required. This chapter outlines the necessary system configurations and tools used during the project lifecycle.

4.2 Hardware Requirements

The hardware requirements depend on the role: whether the system is being used by developers or by end-users (business owners, staff).

4.2.1 For Development Environment

- **Processor:** Intel i5 or equivalent, 2.4 GHz or higher
- **RAM:** Minimum 8 GB
- **Storage:** 256 GB SSD (Recommended)
- **Display:** 13" or higher, Full HD
- **Internet:** Required for library installations and GitHub access

4.2.2 For End-Users

- **Device:** Any device with a modern browser (desktop, laptop, tablet)

- **RAM:** Minimum 2 GB
- **Processor:** Dual-core or higher
- **Browser:** Google Chrome, Mozilla Firefox, or equivalent (latest version)
- **Internet Connection:** Required for accessing the web application hosted on server

4.3 Software Requirements

The following software components were used for development, database management, server hosting, and user interaction:

Software	Purpose
Operating System	Windows 10 or higher / Linux Ubuntu 20.04 or higher
XAMPP (Apache, MySQL, PHP)	Local server setup for running and testing the application
PHP (v7.4 or higher)	Server-side scripting language for back-end logic
MySQL	Relational database management system
phpMyAdmin	Web interface for managing MySQL databases
Visual Studio Code	Source code editor for writing HTML, CSS, JavaScript, and PHP
Web Browser (Chrome, Firefox)	For testing and accessing the web application interface
Bootstrap 5	Front-end framework for responsive design
Git (optional)	Version control tool for tracking code changes
Figma (optional)	Designing wireframes and UI prototypes

Table 4.1: Software Requirements for Inventron

4.4 Minimum vs. Recommended Configurations

Component	**Minimum Configuration**	**Recommended Configuration**
Processor	Dual-core 2.0 GHz	Quad-core 2.5 GHz or above
RAM	4 GB	8 GB or more
Storage	100 GB HDD	256 GB SSD
Operating System	Windows 7/Ubuntu 18.04	Windows 10/Ubuntu 20.04 or higher
Browser	Chrome/Firefox (v90+)	Latest stable version of Chrome/Firefox

Table 4.2: Comparison of Minimum and Recommended System Configurations

Chapter 5

System Design and Architecture

5.1 Introduction

The development of Inventron involved a well-structured system design process to ensure functionality, scalability, and maintainability. This chapter outlines the architectural blueprint and design methodology adopted for developing the inventory management system.

5.2 Design Approach

Inventron follows a modular and layered design strategy, where each component of the application handles a specific responsibility. This separation of concerns enhances maintainability and makes it easier to update or scale individual components without affecting the entire system.

5.2.1 Design Methodology Used

- **Top-down approach:** We started by defining high-level modules such as user authentication, product management, and reporting, then broke them down into detailed sub-modules.
- **Entity-Relationship Modelling:** Used to design the database structure based on real-world business entities.
- **MVC Structure (Model-View-Controller):** Though implemented manually, the system loosely follows MVC for separating business logic, UI, and data.

5.3 System Architecture Overview

The Inventron system is built on a three-tier web architecture consisting of:

- **Presentation Layer:** The front-end UI developed using HTML, CSS, JavaScript, and Bootstrap.
- **Application Layer:** Business logic implemented using PHP, managing the flow of data and user interaction.
- **Data Layer:** MySQL database to store and manage persistent business data (products, users, sales, etc.).

5.4 System Modules

The application is structured into several functional modules:

- **User Authentication:** Login/logout functionality for secure access.
- **Product Management:** Add, update, and delete product information.
- **Inventory Tracking:** Stock level updates and alerts for low inventory.
- **Sales and Revenue Tracking:** Store sales transactions and calculate revenue.
- **Supplier and Customer Management:** Maintain contact and transaction records.
- **Reports and Analytics:** Provide business insights via charts and tables.

5.5 Design Diagrams

This section includes the visual representation of the system design. Please insert your diagrams in the appropriate placeholders below.

5.5.1 System Architecture Diagram

5.5.2 Entity-Relationship Diagram (ERD)

5.5.3 Data Flow Diagram (Level 1)

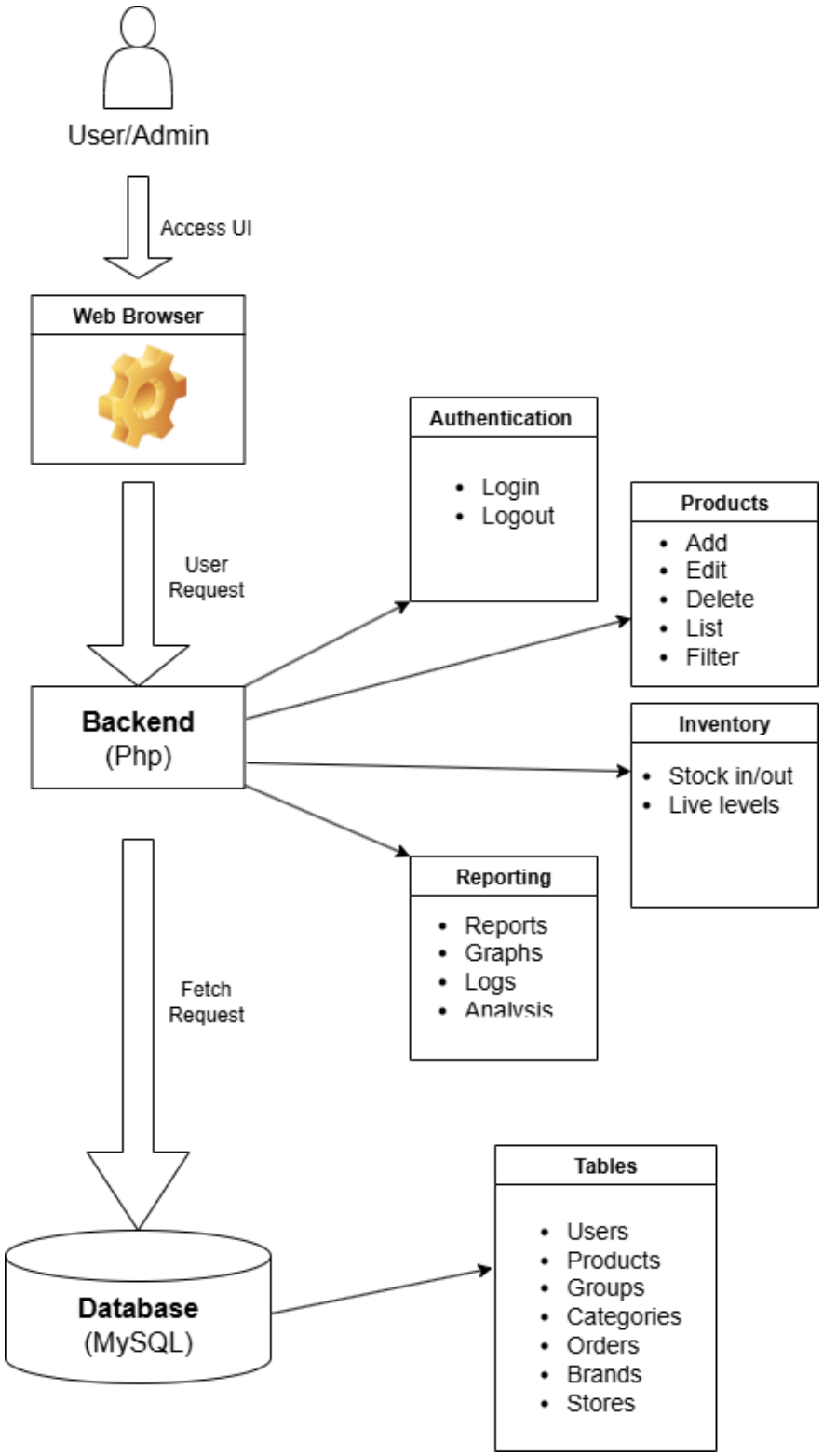


Figure 5.1: System Architecture of Inventron



Figure 5.2: ER Diagram Representing Inventron Database Schema

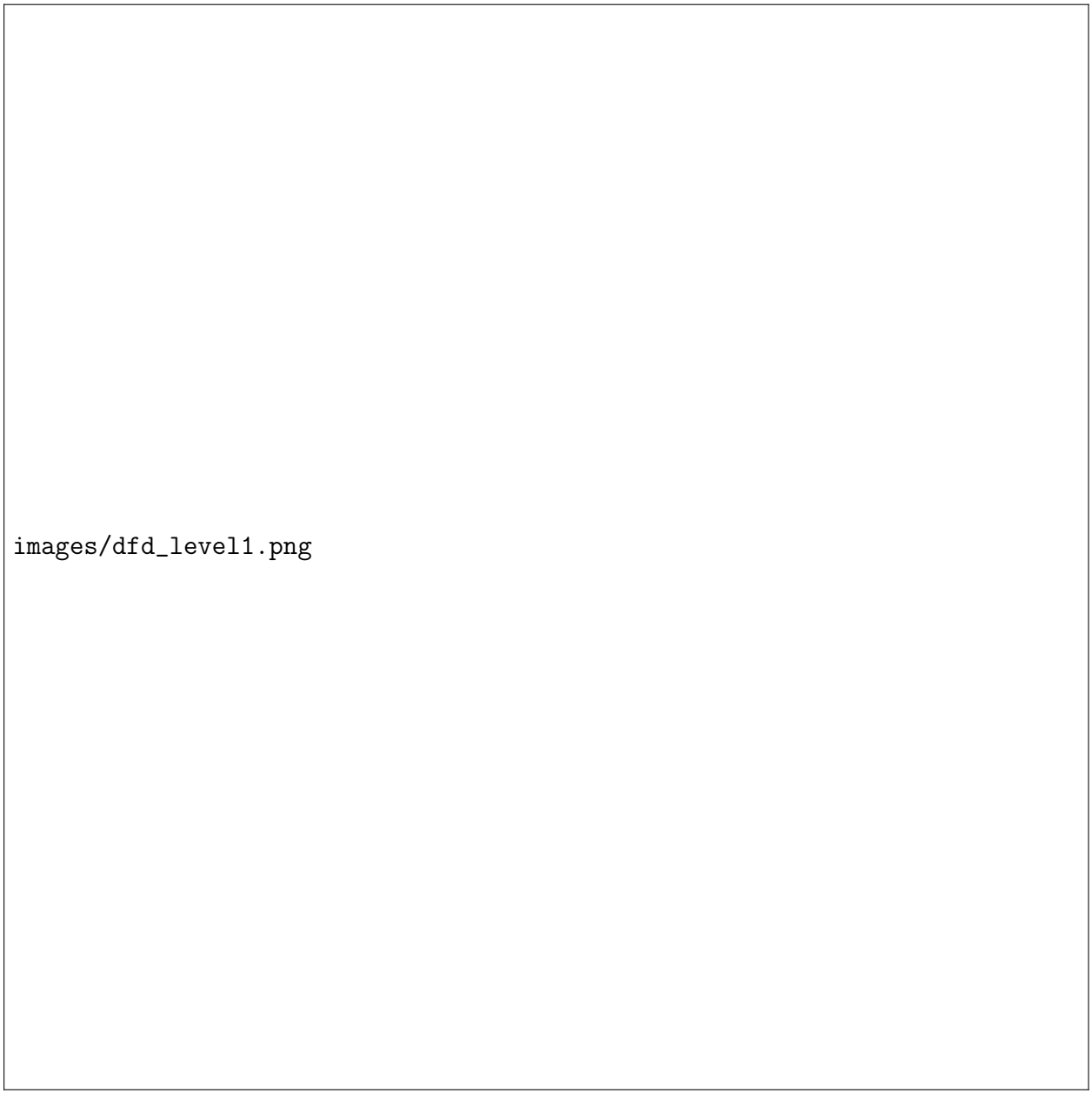


Figure 5.3: Data Flow Diagram – Level 1

Chapter 6

Key Highlights of Inventron

6.1 Overview

Inventron was developed to address the common inventory challenges faced by small and medium-sized enterprises (SMEs). This chapter summarizes the standout features that make Inventron a powerful yet simple inventory management web application.

6.2 User-Friendly Interface

Inventron features a clean and intuitive user interface designed using modern front-end technologies like HTML5, CSS3, Bootstrap, and JavaScript. The layout ensures easy navigation across all modules even for users with minimal technical knowledge. Highlights include:

- Sidebar navigation with clearly labeled sections
- Responsive design that adapts to desktop and mobile browsers
- Consistent color scheme and interactive buttons

6.3 Role-Based Access Control

The application supports different types of users with varied privileges. Each user is assigned a role which restricts or grants access to specific features.

- **Admin:** Full access to all modules (products, users, suppliers, reports)
- **Staff/User:** Limited access (e.g., only add/view stock and sales)
- Secure login with session-based authentication

6.4 Bcrypt-based Authentication

Inventron implements **bcrypt hashing** to securely manage user passwords. This ensures that even if the database is compromised, actual passwords remain safe. Key aspects include:

- Passwords are never stored in plain text
- Each password is hashed using a strong one-way bcrypt algorithm
- Salt is automatically generated and added to each hash to prevent rainbow table attacks
- Verification is done securely by comparing the hash of the entered password with the stored hash

This practice significantly improves user account security and aligns with modern industry standards for web application security.

6.5 Product and Inventory Management

- Add new products with details such as SKU, price, quantity, and description
- Categorize products for better organization (e.g., Electronics, Medicines)
- Automatic update of stock quantities upon sales or purchases
- Alerts for low stock levels to prevent shortages

6.6 Customer and Supplier Records

Inventron allows users to store contact and transactional data of:

- **Suppliers:** Helps track purchasing history and restocking trends
- **Customers:** Stores buyer details, purchase history, and preferences

6.7 Sales Management and Invoicing

- Record sales with product-wise breakdown and total billing
- Auto-deduction of sold quantities from available stock
- Daily/weekly/monthly revenue summaries
- Simple invoicing system to provide receipts to customers

6.8 Analytics and Reporting

- Sales performance charts (bar/line graphs)
- Top-selling products
- Inventory valuation
- Stock history and reorder trends

6.9 Collaboration and Multi-user Capability

- Centralized dashboard for shared business view
- Real-time updates to reflect inventory changes instantly
- Secure multi-session handling

6.10 System Security and Data Integrity

- Input validation to prevent SQL injections and cross-site scripting (XSS)
- Encrypted session tokens
- Strict access controls per user role

6.11 Portability and Accessibility

- Runs on any modern web browser
- No installation required
- Accessible from any device connected to the internet

Chapter 7

User Manual

7.1 Introduction

This user manual serves as a guide to using the core features of the Inventron web application. It is designed for business owners, staff, and administrators who intend to manage inventory, track sales, and monitor stock efficiently using the system.

7.2 System Requirements for Access

To access Inventron, ensure the following:

- A modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge)
- Stable internet connection
- Screen resolution of at least 1366x768 for best visual experience

7.3 Accessing the Application

1. Open any web browser and navigate to the deployed URL of Inventron.
2. Enter your registered email and password to log in.
3. If you are a new user, contact the admin to create an account for you.

7.4 User Roles and Permissions

- **Admin:** Full control of all modules including user creation, product management, and report generation.
- **User/Staff:** Limited access to daily operations such as adding sales or viewing inventory.

7.5 Main Modules and Navigation

Upon successful login, the dashboard will appear with the following main navigation sections in the sidebar:

7.5.1 Dashboard

- Shows key statistics: total products, total sales, total revenue
- Includes graphical representations of sales trends

7.5.2 Products

- Add, edit, delete or view product entries
- Each product includes fields like name, SKU, category, quantity, and price

7.5.3 Suppliers

- Add supplier records including contact details and supply history
- Useful for purchase tracking and reorder management

7.5.4 Customers

- Manage customer details and view past purchases

7.5.5 Sales

- Create new sales transactions
- Automatically updates product quantities

- Generates basic invoices for print/download

7.5.6 Reports

- View sales data filtered by date, product, or customer
- Analyze low-stock items and top-selling products

7.5.7 Settings

- Manage profile details (e.g., password, email)
- Admins can manage user accounts and roles

7.6 Security Tips for Users

- Always log out after use
- Use a strong password (at least 8 characters, with numbers and symbols)
- Never share your credentials with others
- Change your password regularly

7.7 Troubleshooting Common Issues

- **Forgot Password:** Contact admin to reset your account
- **Login Fails:** Check for correct email/password. Clear browser cache if needed.
- **Page Not Loading:** Ensure internet connection or retry after refreshing

7.8 Help and Support

For further assistance, users can reach out to the system administrator or use the provided contact section within the application if enabled.

Chapter 8

Snapshots

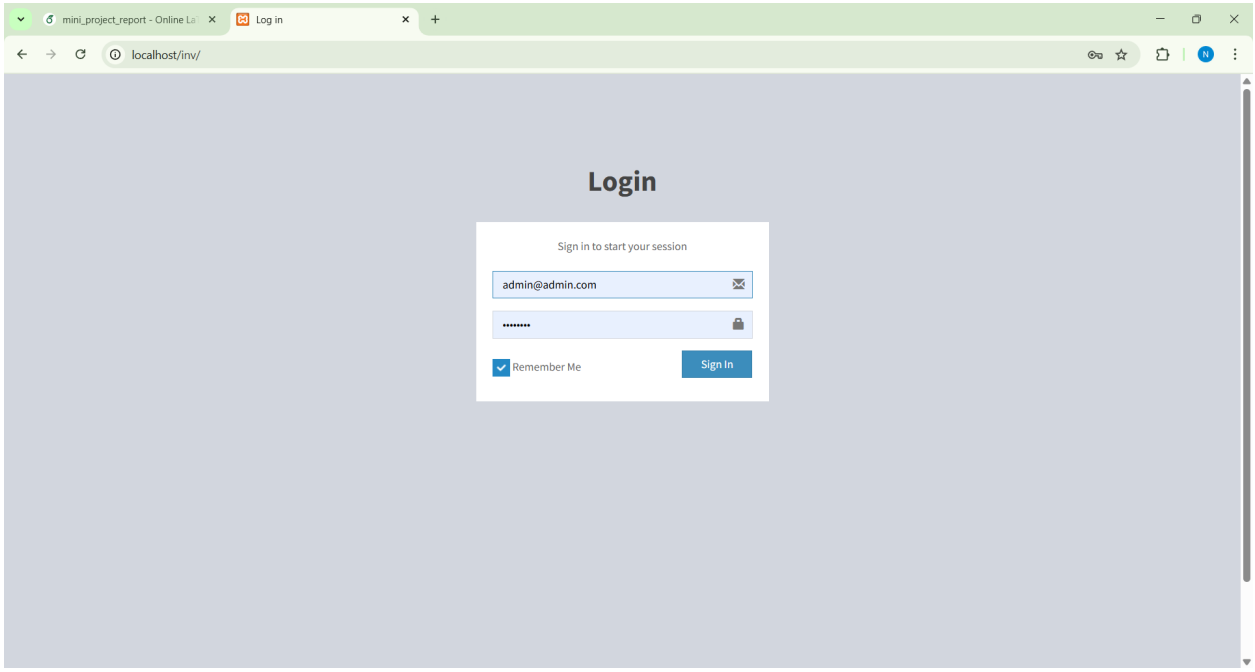


Figure 8.1: Login Page Interface

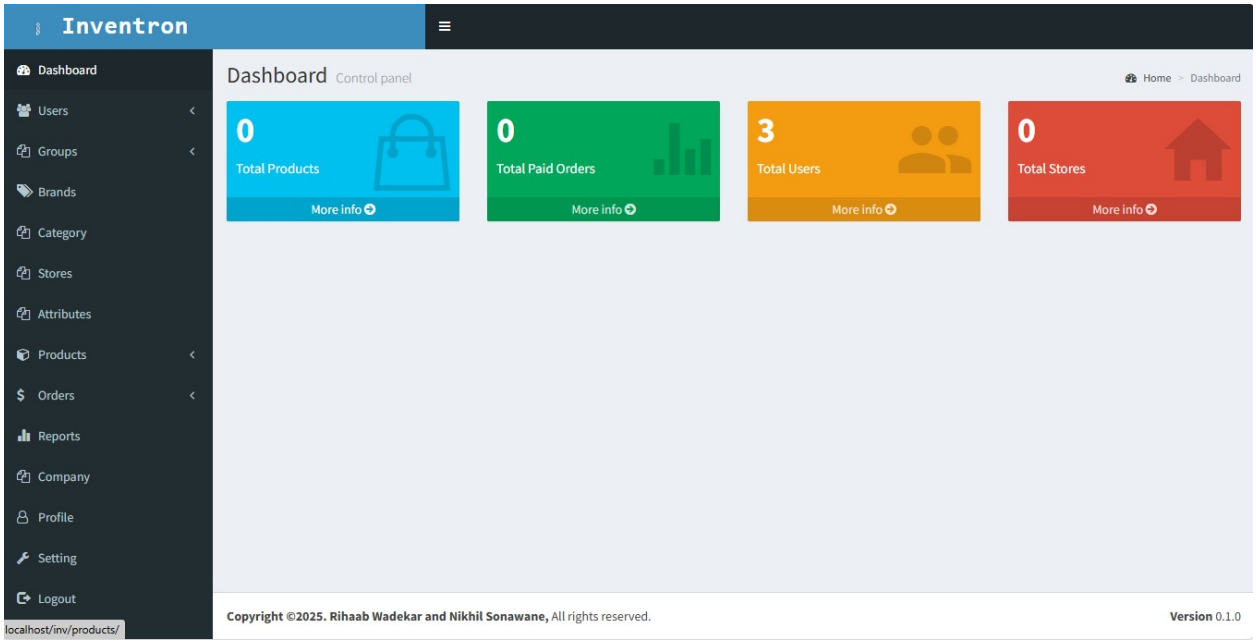


Figure 8.2: User Dashboard

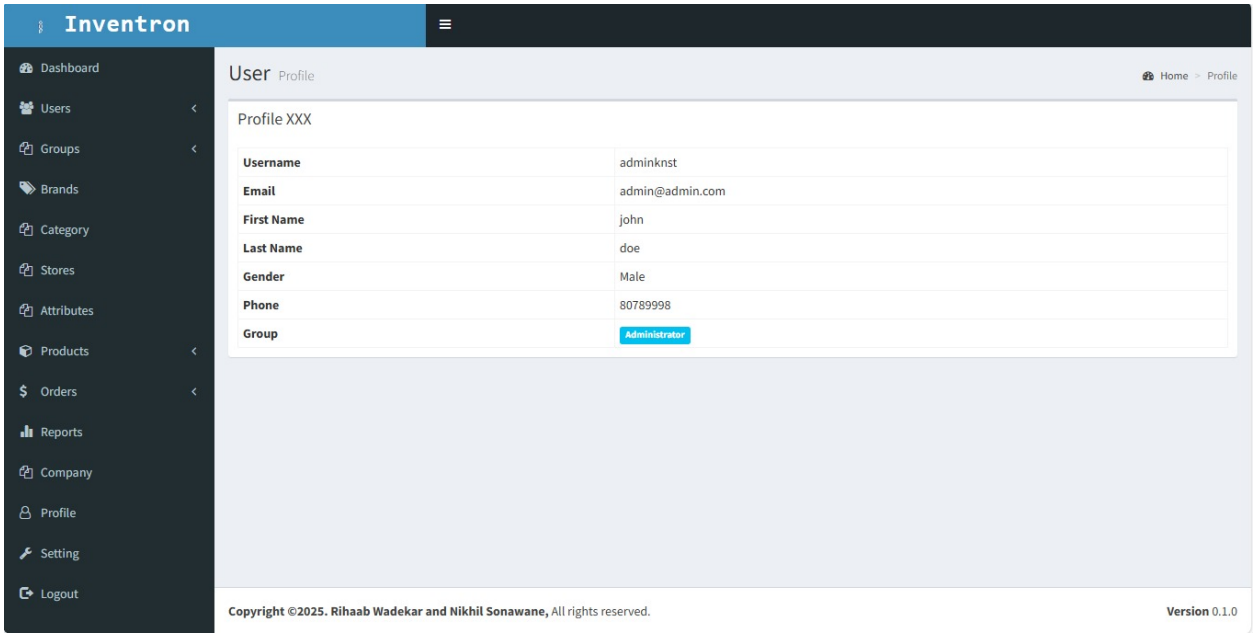


Figure 8.3: Admin Profile

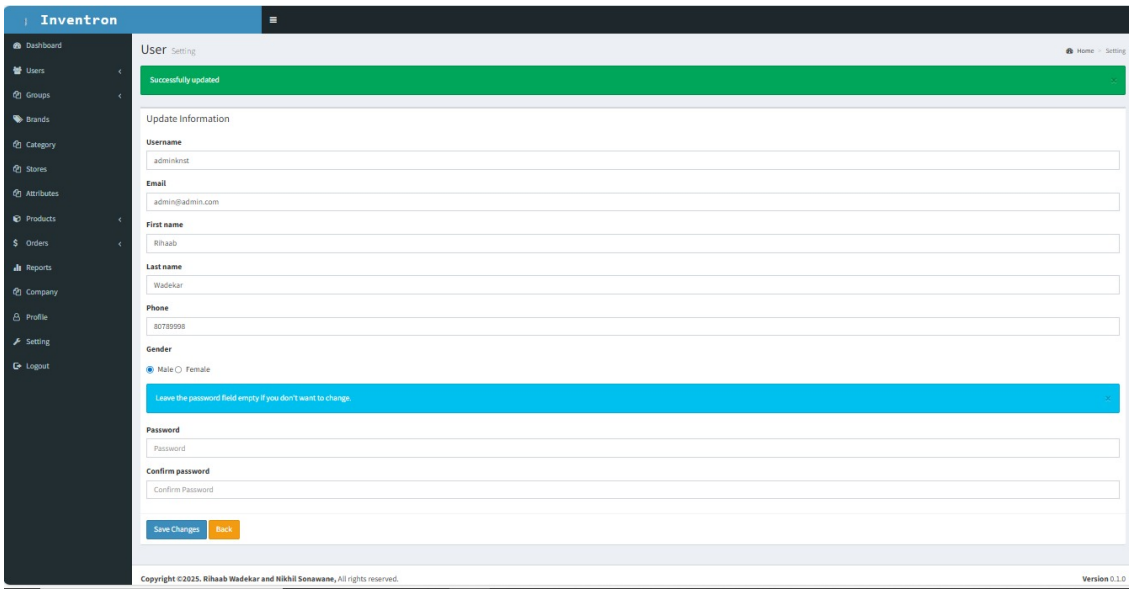


Figure 8.4: User Settings

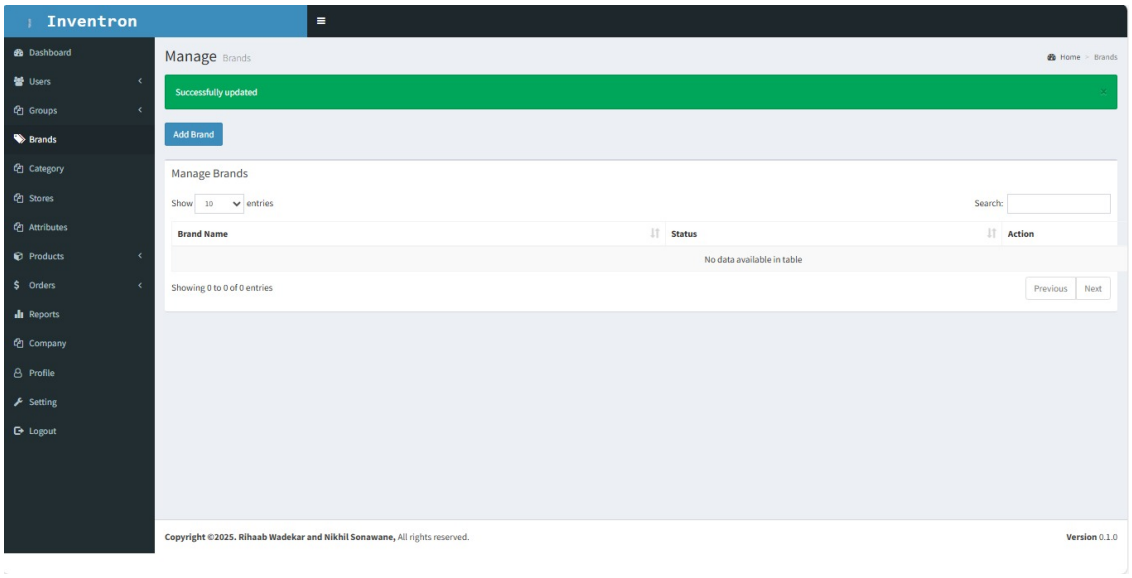


Figure 8.5: Brands management section

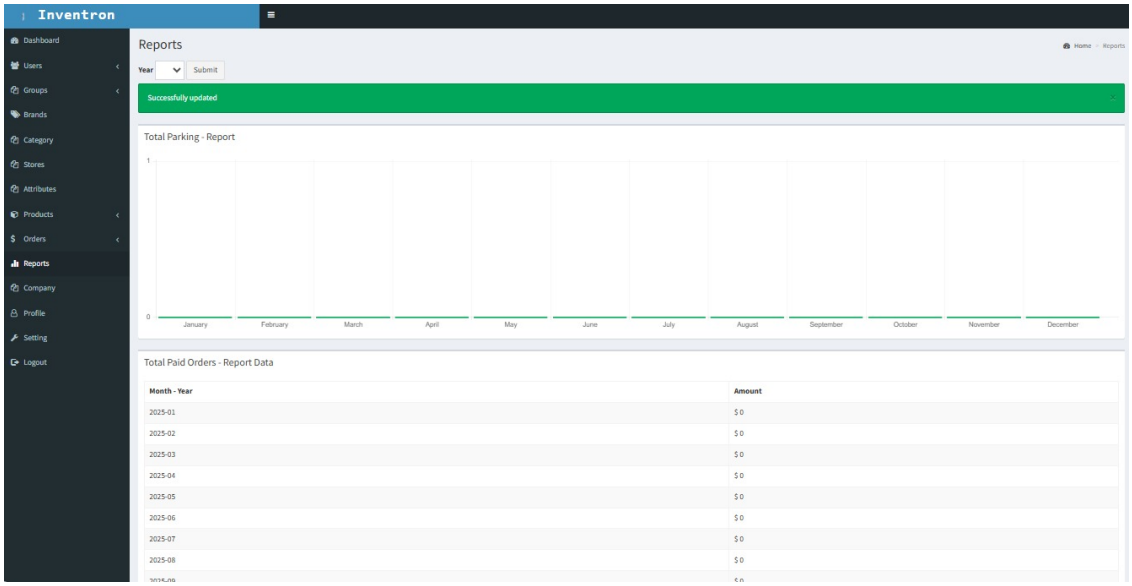


Figure 8.6: Reports

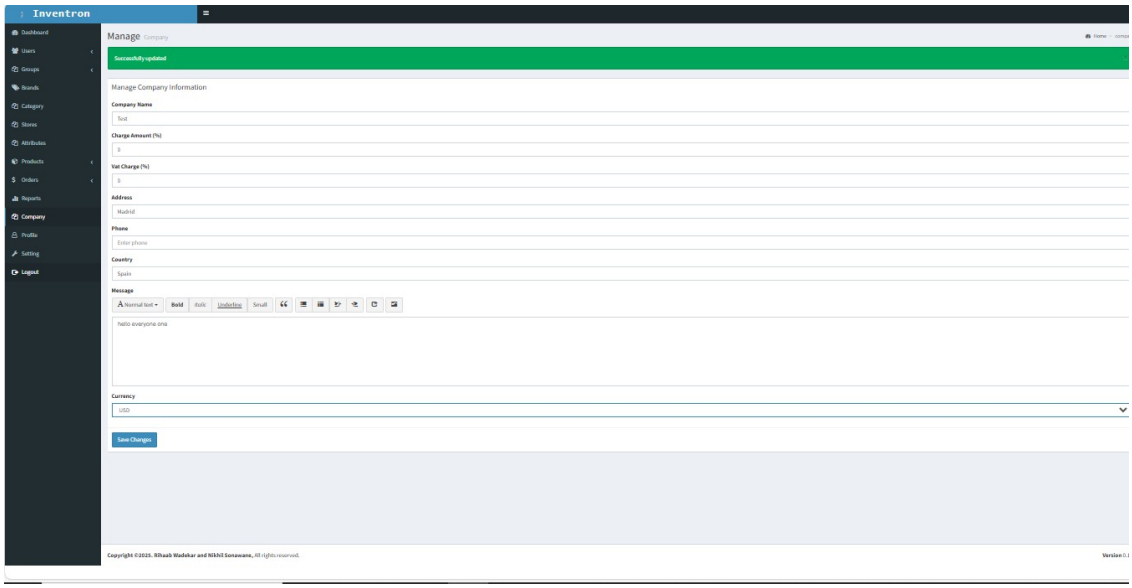


Figure 8.7: Company

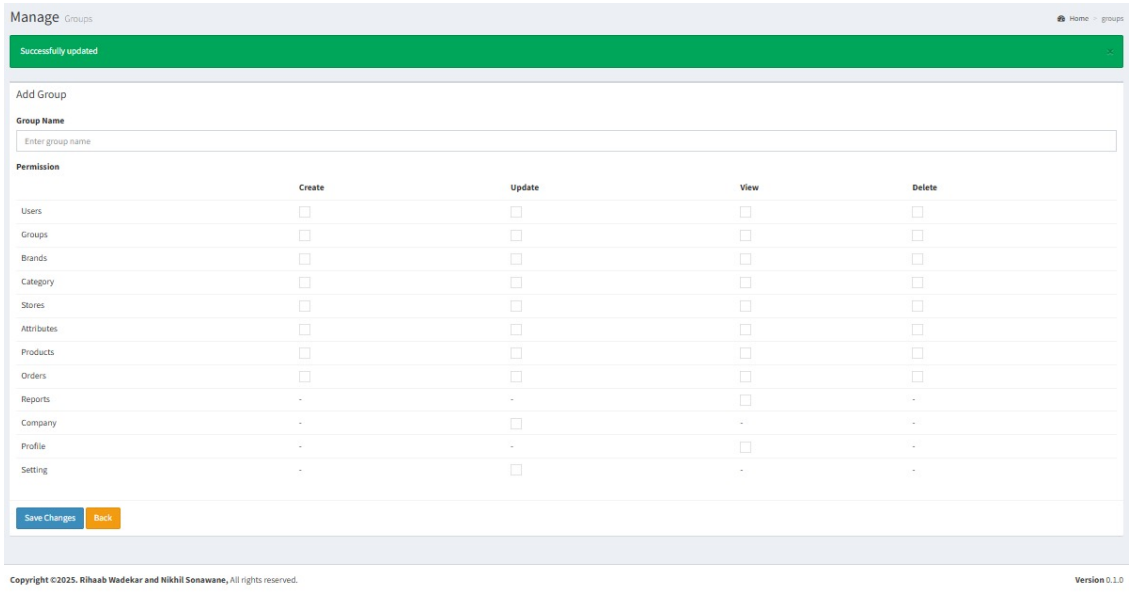


Figure 8.8: Groups

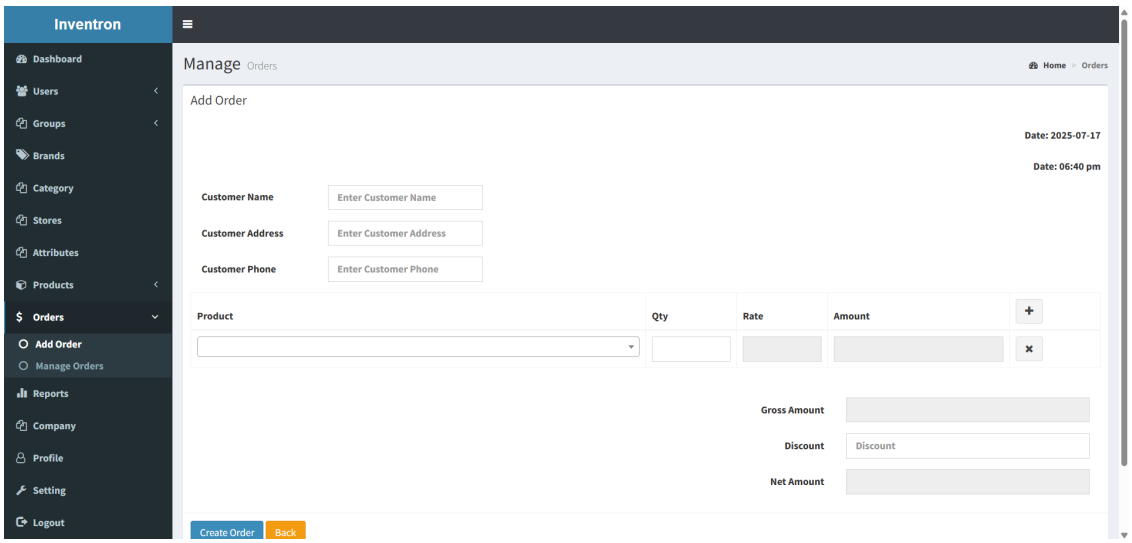


Figure 8.9: Orders

Chapter 9

Conclusion

In today's dynamic business landscape, efficient inventory management is essential for ensuring the profitability, sustainability, and operational fluidity of any organization. The development of the Inventron web application has been a significant step toward simplifying and automating this crucial business function. This project has provided both a practical and academic platform to apply theoretical knowledge in a real-world scenario, merging concepts from software engineering, web development, and database systems.

Inventron successfully offers features like product tracking, sales monitoring, supplier and customer management, and secure authentication, all within an intuitive and responsive user interface. The incorporation of a dashboard with visual analytics empowers users to make informed decisions with ease. The addition of Bycrypted authentication further enhances the system's security by ensuring safe storage of user credentials.

Throughout the development process, focus was given to designing a system that is scalable, modular, and user-friendly. Emphasis on clean UI/UX, backend logic, and database efficiency allowed the application to meet its intended goals. Additionally, feedback-driven iterations helped in refining features and improving overall performance.

The project not only addressed an existing gap in small business inventory solutions but also opened doors for future advancements. The proposed enhancements such as mobile integration, AI-driven forecasting, and cloud synchronization demonstrate the potential of Inventron to evolve into a full-fledged enterprise-grade solution.

In conclusion, this project has been a valuable learning experience in end-to-end software development. It has equipped us with hands-on experience in system design, implementation, testing, and deployment—skills that are highly relevant in the IT industry. Most importantly, Inventron stands as a testament to how technology can be leveraged to solve real-life problems in an efficient and impactful manner.

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